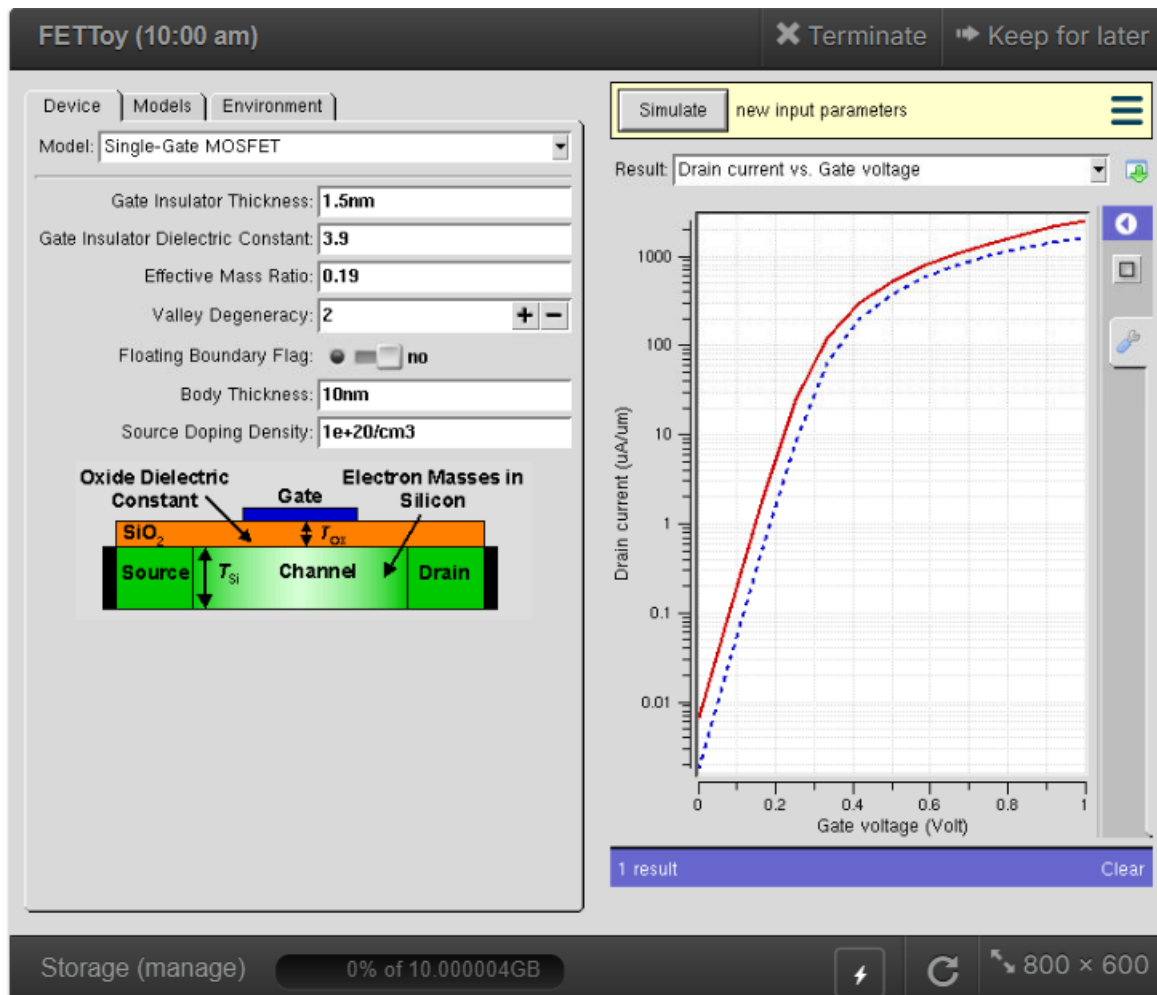
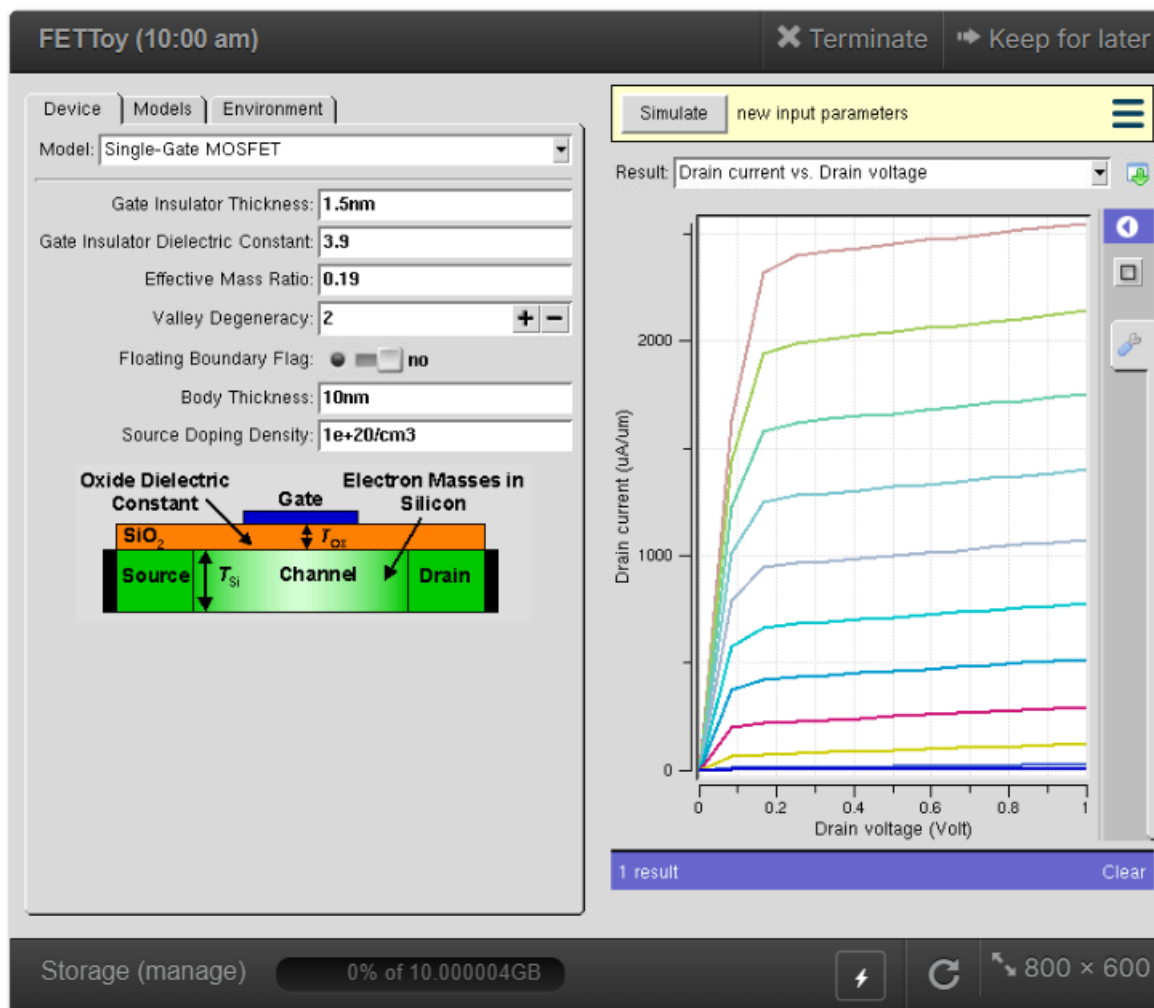


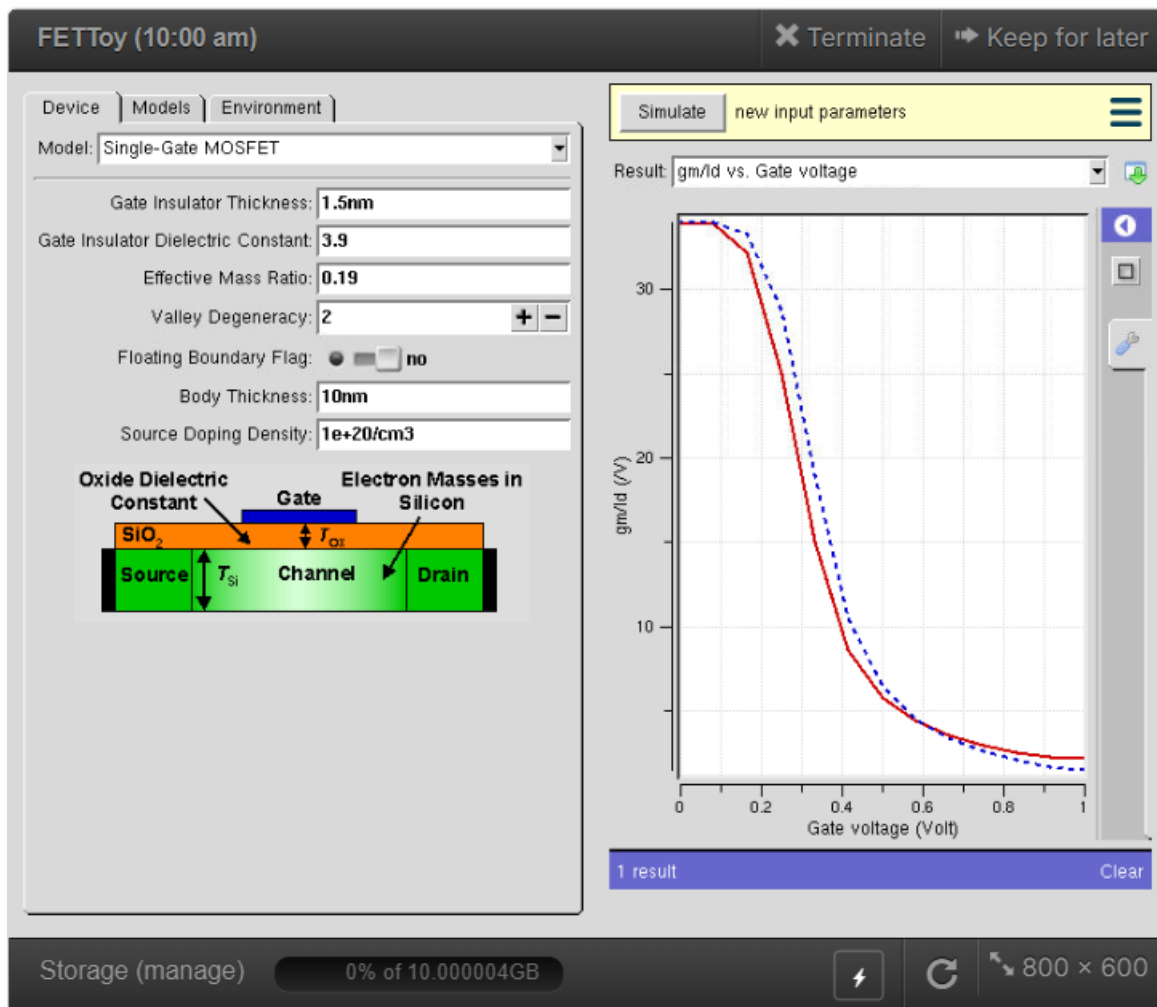
Exp. 7. MOSFET Scaling Analysis and Subthreshold Swing Calculation using NanoHub.



1. This image shows the **FETToy** simulation of a **Single-Gate MOSFET** with a gate insulator thickness of **1.5 nm** and a dielectric constant of **3.9**.
2. The device structure includes the **Source, Gate, Channel, Drain, and SiO₂ gate oxide** layers.
3. The graph displays the **Drain Current (Id) vs. Gate Voltage (Vg)** characteristics of the MOSFET.
4. The drain current increases rapidly with gate voltage, indicating effective channel formation and improved electron conduction.
5. This simulation is useful for analyzing and optimizing **MOSFET performance in CMOS circuits, integrated circuits, and low-power nanoelectronic devices**.



1. This image shows the **FETToy** simulation of a **Single-Gate MOSFET** displaying the **Drain Current (I_d) vs. Drain Voltage (V_d)** characteristics.
2. The graph contains multiple curves representing the drain current for **different gate voltage (V_g) values**.
3. The drain current increases rapidly with drain voltage and then reaches **saturation**, which is a typical MOSFET behavior.
4. Higher gate voltages produce **higher drain currents** due to stronger channel formation and increased carrier flow.
5. This simulation is useful for evaluating **MOSFET output characteristics** and optimizing **CMOS circuits, integrated circuits, and nanoelectronic devices**.



1. This image shows the **FETToy** simulation of a **Single-Gate MOSFET** displaying the **Transconductance (gm/Id) vs. Gate Voltage (V_g)** characteristics.
2. The graph illustrates how the **gm/Id ratio changes** as the gate voltage increases.
3. The **red and blue curves** compare the simulated transconductance performance under different conditions.
4. The **gm/Id value decreases** with increasing gate voltage, indicating reduced transconductance efficiency at higher operating voltages.
5. This simulation is useful for optimizing **MOSFET design, low-power CMOS circuits, analog integrated circuits, and nanoelectronic devices**.
- 6.